

You are a proficient Java programmer and familiar with various algorithms.

Your task is to implement a Java code given a natural language description illustrating an algorithm and a Java signature.

On a line, colliding robots battle. The stronger survives with one less health, eliminating the weaker. If healths are equal, both are eliminated

Please implement the function with the following template.

```
java ⌵ Collapse Copy  
1 class Solution {  
2     public List<Integer> survivedRobotsHealths(int[] positions, int[] healths, String directions)  
3     }  
4 }
```

HKGAI-V1

-- Free credit: 8.0; Paid credit: 0.0 --

```
java ⌵ Collapse Save Copy  
1 import java.util.*;  
2  
3 class Solution {  
4     public List<Integer> survivedRobotsHealths(int[] positions, int[] healths, String  
5         int n = positions.length;  
6         Integer[] indices = new Integer[n];  
7         for (int i = 0; i < n; i++) {  
8             indices[i] = i;  
9         }  
10        Arrays.sort(indices, (a, b) -> positions[a] - positions[b]);  
11  
12        Stack<Integer> stack = new Stack<>();  
13        for (int i : indices) {  
14            if (directions.charAt(i) == 'R') {  
15                stack.push(i);  
16            } else {  
17                while (!stack.isEmpty() && healths[i] > 0) {  
18                    int top = stack.peek();  
19                    if (healths[top] > healths[i]) {  
20                        healths[top]--;  
21                        healths[i] = 0;  
22                    } else if (healths[top] < healths[i]) {  
23                        healths[i]--;  
24                        healths[top] = 0;  
25                        stack.pop();  
26                    } else {  
27                        healths[top] = 0;  
28                        healths[i] = 0;  
29                        stack.pop();  
30                    }  
31                }  
32            }  
33        }  
34        List<Integer> result = new ArrayList<>();  
35        for (int i : indices) {  
36            if (healths[i] > 0) {  
37                result.add(i);  
38            }  
39        }  
40        return result;  
41    }  
42 }
```

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java

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```
1 import java.util.*;  
2  
3 class Solution {
```